Tracing Antiquity: References to Greco-Roman Authors in Modern Academic Discourse

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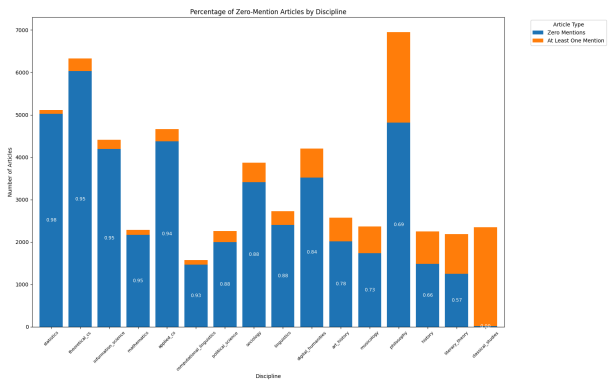
Introduction

The Latin term “ canon” began to be used in the 18th century to refer to selective lists of ancient authors. In C lassics, “canon” and “canonical” are still used to refer to lists of authors who have been selected since antiquity according to many different criteria (Pfeiffer 1968; Nicolai 2007). Given that scholarship plays a key role in shaping the canon, we consider a corpus of contemporary academic articles from different disciplines and use a new digital approach to retrieve and detect references to Greco-Roman authors. These references may signal authority and belonging to intellectual tradition, while their absence or selective use may reflect shifts in disciplinary priorities. The aim of this paper is to discover how ancient authors are represented in the scholarly discourse of different fields, and whether the canon shaped by classical scholarship is consistent across disciplines. This research is part of the MECANO project, which explores “the mechanics of canon formation and knowledge transmission of ancient Greek and Latin texts”. [1] This topic is central to literary studies, and digital technologies are proving to be important in tracing the presence or absence of authors in the evolution of the canon (Romanello 2013; Netz 2020; Berti 2024).

Corpus and Methods

Corpus creation

We use papers published in journals across various disciplines as a proxy for academic discourse. This approach offers a focused perspective, but represents only one facet of the broader topic. Future research may also include other forms of scholarly communication, e.g., books, newspapers, and online forums. The aim of this paper is to demonstrate that even simple digital methods can reveal intriguing patterns in referencing strategies across disciplines. We re-use an existing corpus of academic papers published in a scientometric study by Luhmann / Burghardt (2022). Our corpus comprises a total of 56,116 articles with over 299 million tokens. It covers a timeframe of three decades (1990-2019) and features articles from 16 disciplines (Fig. 1).



Overview of disciplines and number of articles per discipline without references (blue) or with at least one reference (orange) to our selected authors.

Detection of ancient authors

References to Greco-Roman authors take the form of mentions of their name s or their works, direct or modified quotations, translations, and nuanced allusions to foundational texts. T he use

of abbreviations is also common in Classics . However, this study will focus on detecting author names. We thus use a dictionary containing 200 names of the most important ancient authors [2] . The selection of authors results from a manual mapping between the Trismegistos Author gazetteer and the “L'Année Philologique” (APh) database [3] . The former is a collection of over 10,000 authors from antiquity. The latter is a bibliographic database of scholarly publications about authors and aspects of the a ncient Greek and Roman w orld, such as literature, language, history, and culture. Founded in 1924 by Jules Marouzeau, it still is today a fundamental reference resource for researchers in Classics , with extensive bibliographic records of books, journal articles, and other scholarly work s .

The initial dictionary contained only one spelling variant for each author. To address this issue, the Wikidata Query Service was employed to incorporate the names in a range of variants and languages (e.g., Seneca the Younger, Lucius Annaeus Seneca, Seneca, Sénèque). This complete dictionary was the vocabulary to perform a systematic keyword search in our corpus using the CountVectorizer function of scikit-learn. The occurrences of various aliases were counted. The aggregated totals for aliases corresponding to the same author were subsequently calculated [4] . We observed that variations in language (e.g., “Epicurus”, “Epicure”) were more prevalent than demonyms (e.g., “the Stagyrite”), appellatives (e.g., “father of comedy” for Aristophanes), or longer denominations in general. Additionally, we corrected the counts to address the issue of long denominations containing shorter ones.

Results and discussion

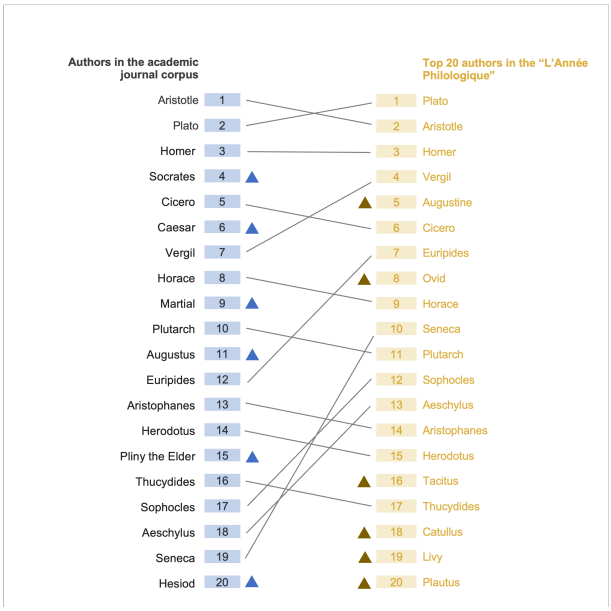
Distribution of references across disciplines

Using our enhanced dictionary, we found a substantial total of 244,748 references spread over 36,787 articles. Interestingly, the percentage of articles with at least one mention to an ancient author is at least 2-5 % for all the more technical and scientific disciplines (Fig. 1). This is perhaps unexpectedly high given the perceived distance between contemporary sciences and the epistemological questions and approaches of antiquity.

Philosophy, history, and literary theory exhibit fewer zero-mention articles, reflecting greater engagement with the classical canon. Unsurprisingly, Classical Studies has no zero-mention articles. This visualisation highlights the varying integration of canonical works across disciplines, revealing stark contrasts between the sciences and the humanities.

Most referenced authors

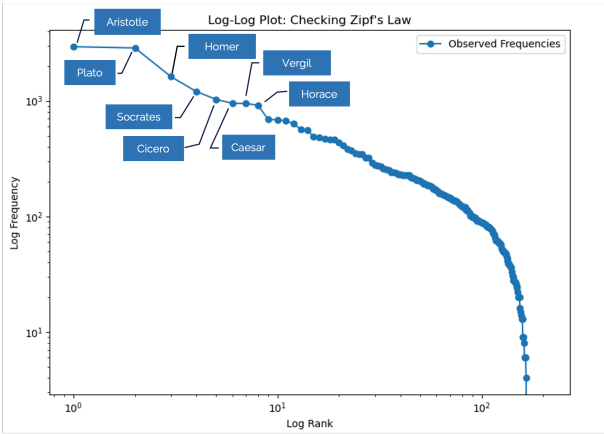
In the following, we will employ a binary marker to indicate the presence of each author in an article: either not cited (0) or cited at least once (1). Fig. 2 compares the 20 most frequent authors in our corpus with the top 20 from APh. Notable similarities include Aristotle, Plato, and Homer occupying the top three ranks in both corpora. To get a more objective measure of the similarity of both rankings, we conducted a Spearman rank correlation analysis, which resulted in a weakly positively correlated score (0.4346, p ~ 0.056). The weak positive correlation suggests that there is some agreement between the two corpora about which authors are important, but it's not a strong agreement.



Comparison of rankings of our corpus and APh.

Interestingly, Seneca is distinguished by the greatest discrepancy between the two top 20 rankings. Some divergences can be attributed to factors beyond authorship: Socrates, for example, is more recognised as a non-author philosopher, while Caesar and Augustus – as political figures of the Roman Empire – are cited for historical reference. They would therefore be expected to feature more prominently in a diverse corpus than in a ranking of Classical Studies. The absence of

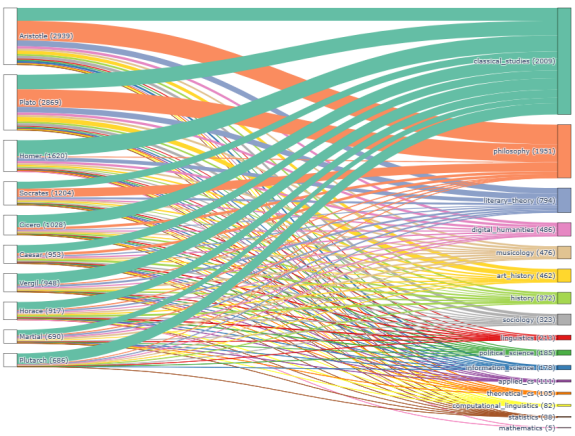
Augustine may reflect the declining influence of Christianity, as evidenced by the complete lack of Christian authors in our top 20. Examining the distribution of authors beyond the top 20 reveals a typical Zipf distribution, where a small number of authors account for the majority of references in the corpus (Fig. 3).



Observed frequencies of ancient author references vs. rank (log-log scale).

Volume of references by discipline

Subsequently, we analyse the distribution of frequently referenced authors across disciplines (Fig. 4). Classics has the highest concentration of references, while Digital Humanities has a volume comparable to other humanities disciplines. In contrast, Mathematics references a limited set of authors, such as Ptolemy and Euclid, rather than those prominent in other fields.



Flow chart depicting citation distributions of authors across various disciplines.

Disciplinary citation strategies and correlations

To compare the referencing patterns per discipline, Spearman rank correlations are again calculated between all disciplines, the corpus-wide author ranking and the APh ranking. Ten comparisons are found to be statistically significant (see Fig. 5). Negative correlations indicate that items displayed high in one ranking are low in the other.

Discipline 1	Discipline 2	Spearman corr	p-value
History	L'Année Philologique	0,8303	0,0029
Political Science	Sociology	0,7576	0,0111
Art History	Information Science	-0,7333	0,0158
Musicology	L'Année Philologique	0,7091	0,0217
Classical Studies	Sociology	0,697	0,0251
History	Political Science	-0,697	0,0251
Political Science	L'Année Philologique	-0,697	0,0251
Art History	Computational Linguistics	0,6848	0,0289
Information Science	Philosophy	0,6606	0,0376
Computational Linguistics	Theoretical Computer Science	-0,6485	0,0425

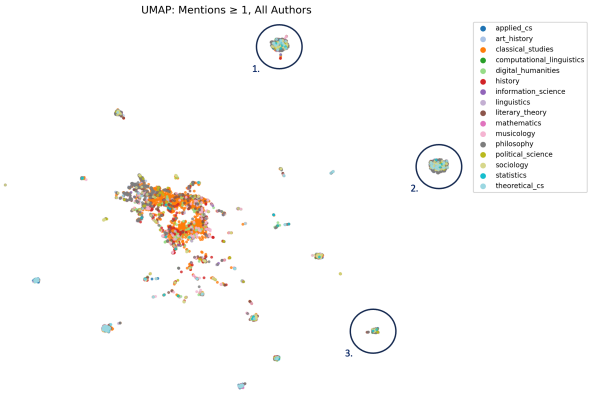
Tabular view of Spearman correlations with 95% confidence level.

The strong correlations between apparently dissimilar disciplines, such as Classics and Sociology, could be explained by the Zipfian distribution: top authors dominate citations overall, and this pattern holds within each discipline. While we might expect close disciplines to correlate strongly, this is only true for Political Science and Sociology. Through close reading, we identified several citation strategies in our corpus. Some authors are cited for analogies, such as Plato in scientific disciplines. Others are compared, like Protagoras and Socrates in Political Sciences. Canonical authors in Mathematics, such as Ptolemy and Euclid, are referenced for their algorithms, not for their treatises on geometry.

The construction of a UMAP to visualise citation data does not reveal any evidence of distinct citation strategies across disciplinary boundaries, as no clusters of a single colour are visible (Fig. 6). However, some clusters show notable citation patterns. For instance, Plato, Aristotle and Socrates are individually cited in clusters 1, 2 and 3 respectively. This highlights the importance of these particular authors, who are often the only reference to antiquity in many articles, regardless of discipline. The citation patterns in Classical Studies, represented by the dense central cluster, are too complex to study in this manner. Further research could include network analysis to examine co-citation patterns and the likelihood of a second author being cited after the first.

Conclusions

The results reveal diverse strategies for citing Greco-Roman authors across academic disciplines. While authors like Plato are employed analogically in scientific contexts, others, such as Caesar and Augustus, serve as historical references, reflecting notable divergences from their rankings in L'Année Philologique. The data exhibit a Zipfian distribution, with a small subset of authors dominating citations across the corpus, while disciplinary clustering, observed through UMAP analysis, highlights distinct yet overlapping citation strategies. Despite the limited temporal scope of the corpus and the challenges of keyword search—particularly with disambiguation and loss of context—these findings highlight the enduring influence of Greco-Roman authors and their evolving roles in contemporary scholarship. They suggest the need for advanced methods and expanded corpora to better understand citation dynamics and canonisation processes. Future research will adopt state-of-the-art, ML-based reference mining methods (Hiltmann et al., 2022).



UMAP of the citation information and 3 clusters with clear citation strategy. Articles with 0 mentions to our authors were removed, as they distorted the clustering.

Notes

1.
2. Dictionary with 200 most frequent authors as they occur in "L'Année Philologique": csv
3. and respectively.
4. Mapping dictionary of authors with Wikidata ID and their related aliases: https://github.com/luisarip/DH2025/blob/main/aliases.csv

Bibliography

Berti, Monica (2024): "Digital Practice for Studying the Indirect Transmission of Classical Authors and Works", in: Favi, Federico / Mastellari, Virginia (eds.): *Treasuries of Literature: Anthologies, Lexica, Scholia and the Indirect Tradition of Classical Texts in the Greek World*. Berlin: De Gruyter, 189–206. DOI: 10.1515/9783111386010-010.

Hiltmann, Torsten / Dröge, Martin / Dresselhaus, Nicole / Grallert, Till / Althage, Melanie / Bayer, Paul / Eckenstaler, Sophie / Mendi, Koray / Schmitz, Jascha Marijn / Schneider, Philipp / Sczeponik, Wiebke / Skibba, Anica (2025): "NER4all or Context is All You Need: Using LLMs for Low-Effort, High-Performance NER on Historical Texts. A Humanities Informed Approach". DOI: 10.48550/arXiv.2502.04351.

Luhmann, Jan / Burghardt, Manuel (2022): "Digital Humanities—A Discipline in Its Own Right? An Analysis of the Role and Position of Digital Humanities in the Academic Landscape", in: *Journal of the Association for Information Science and Technology* 73, 2: 148–171. DOI: 10.1002/asi.24533.

Netz, Reviel (2020): *Scale, Space and Canon in Ancient Literary Culture*. Cambridge: Cambridge University Press. DOI: 10.1017/9781108686945.

Nicolai, Roberto (2007): "Il canone tra classicità e classicismo", in: *Critica del testo* 10, 1: 95–103. DOI: 10.1400/118571.

Pfeiffer, Rudolf (1968): *History of Classical Scholarship from the Beginnings to the End of the Hellenistic Age*. Oxford: Clarendon Press.

Romanello, Matteo (2013): "Creating an Annotated Corpus for Extracting Canonical Citations from Classics-Related Texts by Using Active Annotation", in: Gelbukh, Alexander (ed.): *Computational Linguistics and Intelligent Text Processing*. CICLing 2013. Lecture Notes in Computer Science, vol. 7816. Berlin: Springer, 83–94. DOI: 10.1007/978-3-642-37247-6_6.